ASK//WHY THE UNASKED QUESTION ENGINE

Build report: *what was built, tested and fixed.*

A working prototype that analyses project documents, datasets and workflows to find hidden assumptions, contradictions, missing constraints and unanswered questions. Every cited excerpt is checked against the source text.

49	5	46 / 46	24
automated tests passing	file formats parsed	sample citations verified	findings on a live upload

01 Summary

The project folder started empty: no code and no git repository. ASK//WHY was built from scratch. It has a React frontend, a Python FastAPI backend, a rule-based analysis engine that needs no AI, an optional Claude layer, a synthetic sample investigation that works with no API key, and automated tests.

> **Status.** The application runs locally at `http://localhost:5174` (start it with `./start.sh`). The full demo was tested end to end in headless Chrome with no console errors. The one part not tested against the live API is the AI layer, because no API key was available (see section 07).

02 Technology

LAYER	STACK
Frontend	React, TypeScript and Vite; Tailwind CSS with custom design tokens; React Flow for the evidence graph; Recharts for the impact × confidence chart
Backend	Python FastAPI, Pydantic schemas, SQLite storage
File reading	PyMuPDF (PDF), python-docx (DOCX), pandas (CSV), openpyxl (XLSX)
AI (optional)	Claude `claude-opus-5-5` with typed output, called only from the backend; the API key never reaches the browser
Design	"Investigative editorial" style: charcoal canvas, warm off-white document surfaces, amber signal colour, restrained red; Fraunces, Inter Tight and JetBrains Mono fonts

```
upload → validate → parse → evidence units (stable IDs, e.g. S2.p3.04, S3.r0017) → claims & assumptions →
cross-source comparison → gaps & failure scenarios → findings (rules [+ Claude]) → citation check → scoring → graph & scenarios
```

03 Features delivered

FILE INGESTION

PDF keeps page numbers; DOCX keeps paragraphs, headings and tables; TXT/MD keep line numbers; CSV/XLSX keep sheet names and row numbers. Unsupported, empty or oversized files are rejected with a clear message.

BLIND SPOT RADAR

Flags unsupported assumptions (stated or implied), dependencies with no fallback, missing topics (security, testing, ownership), and high-impact open questions.

CONTRADICTION LAB

Finds conflicting dates, budgets, prices, headcounts and requirements across files. Shows both claims side by side with a verdict (true / possible / explainable) and what would resolve it. It never picks a side.

DATASET AUDIT

Missing values, duplicate rows, outliers, placeholder values such as 9999, inconsistent labels and out-of-range values, each with row references.

CITATION CHECKING

An excerpt is verified only if it really appears in the extracted text. Paraphrases are marked unverified in red, lower the finding's confidence, and are kept out of the graph.

TRANSPARENT SCORING

Impact and confidence come from documented, weighted factors. Each finding shows its score breakdown; a slider re-weights the ranking; reviewers can override impact.

EVIDENCE CONSTELLATION

An interactive graph: sources → cited passages → assumptions / constraints / risks → questions → actions. Clicking a finding traces its evidence.

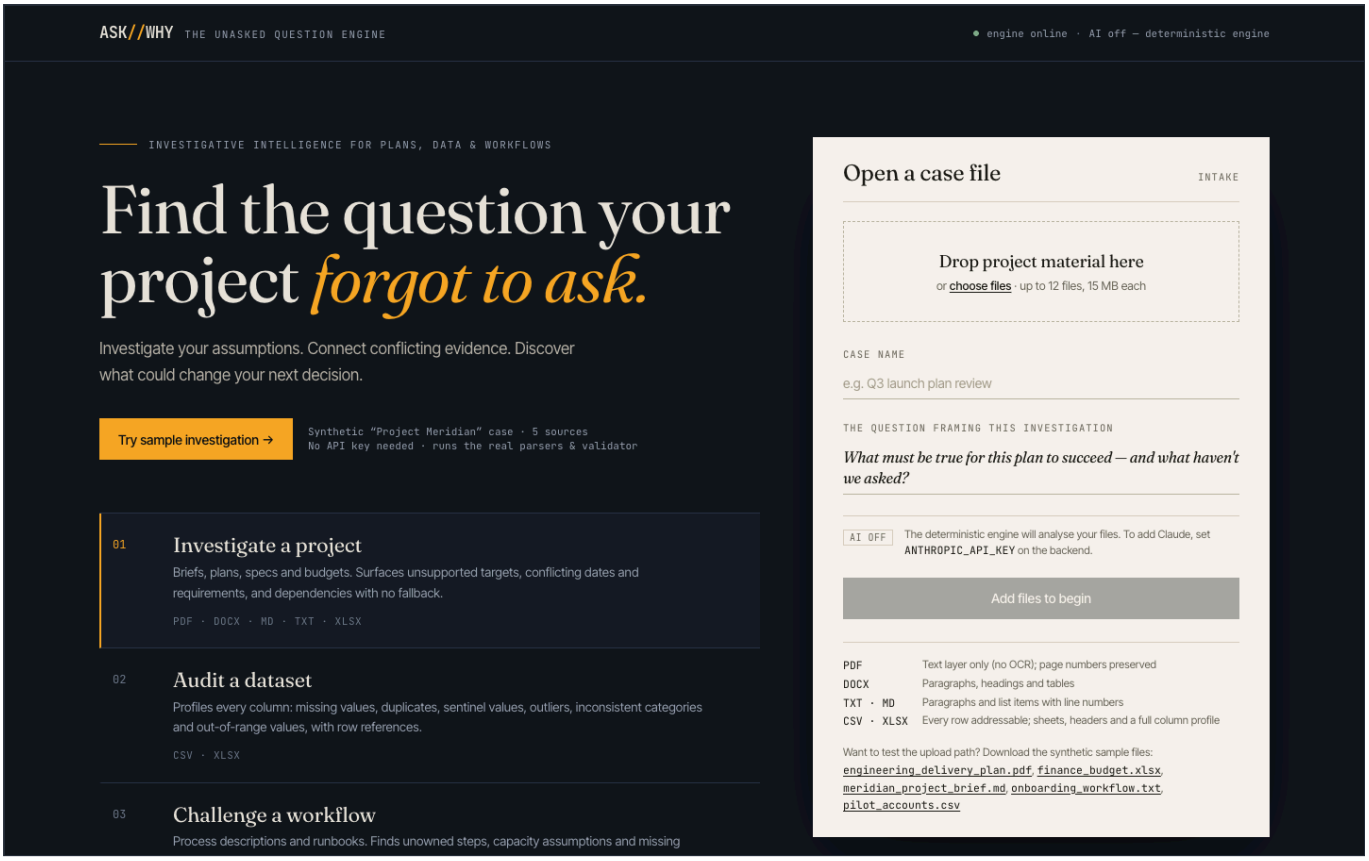
DECISION SIMULATOR

"What if acquisition is 50% slower?" Arithmetic on cited figures is kept separate from evidence-backed conclusions and from clearly labelled hypothetical outcomes.

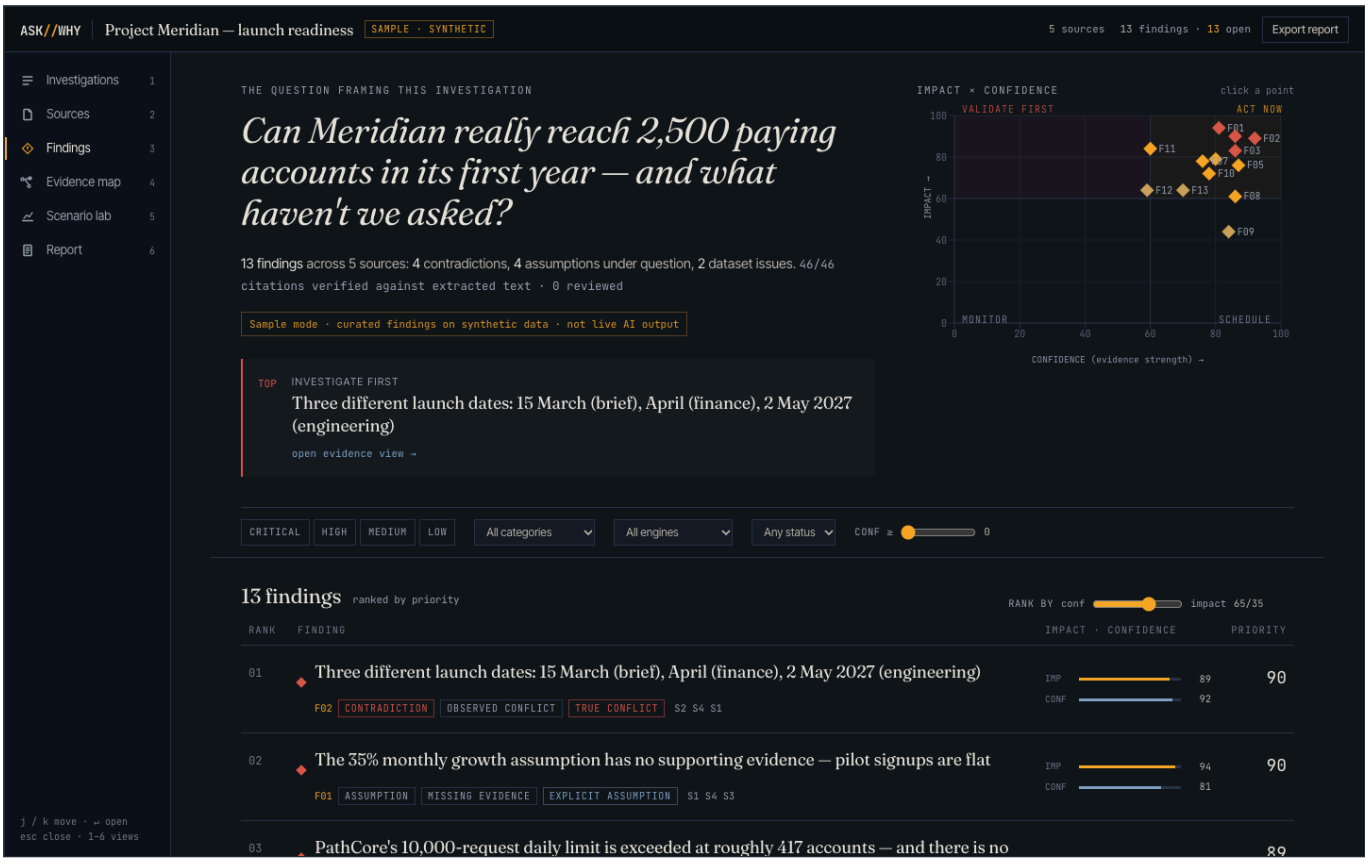
Review workflow, reports and resilience

- Mark findings validated, resolved or dismissed, and add reviewer notes. The state is saved.
- Printable report (Print / Save as PDF) and a Markdown download.
- The sample investigation works with no API key, and still works when the backend is down, using a copy bundled into the frontend.
- Keyboard shortcuts (1–6 views, j/k, Enter, Esc), reduced-motion support, and pages that work at phone width.

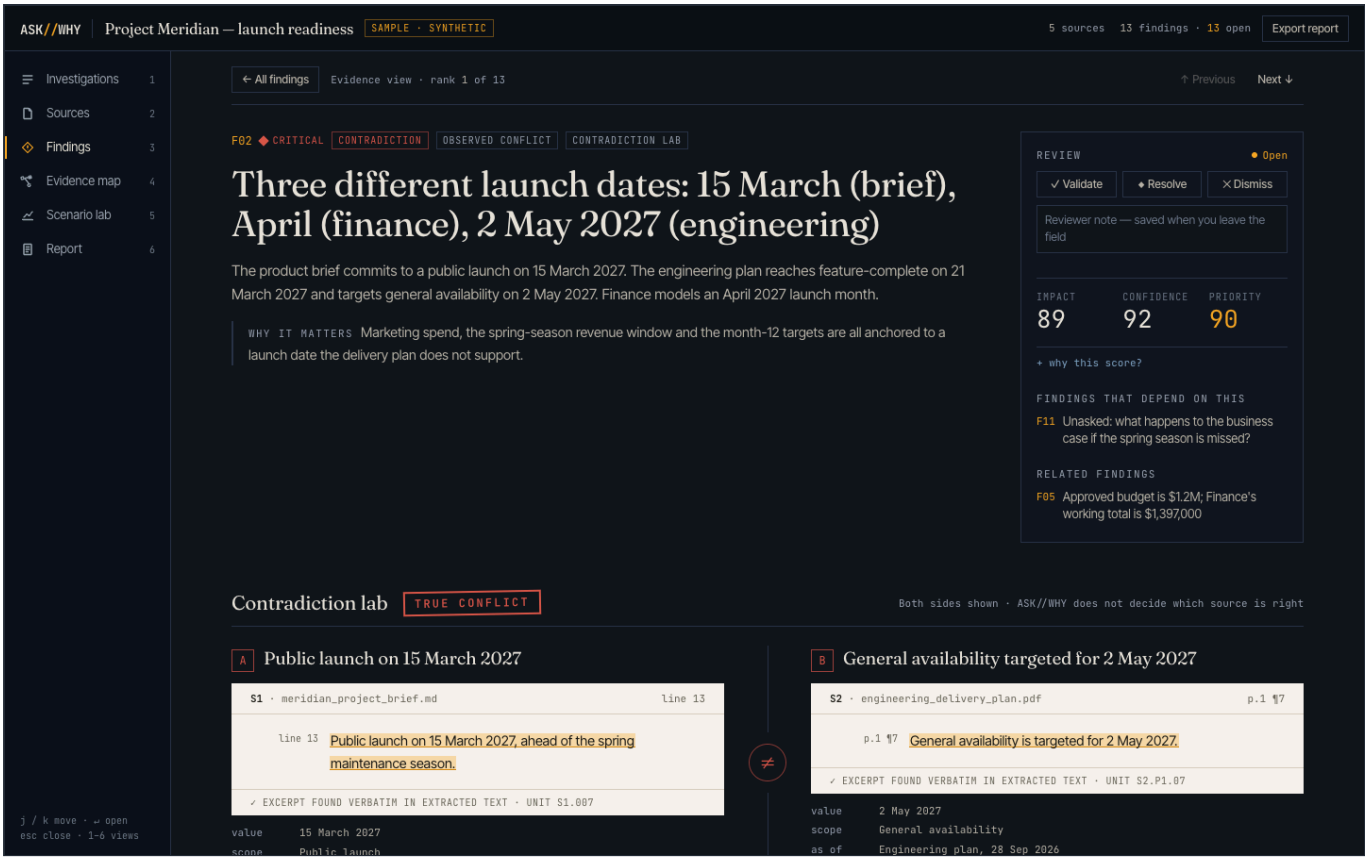
04 The product, screen by screen



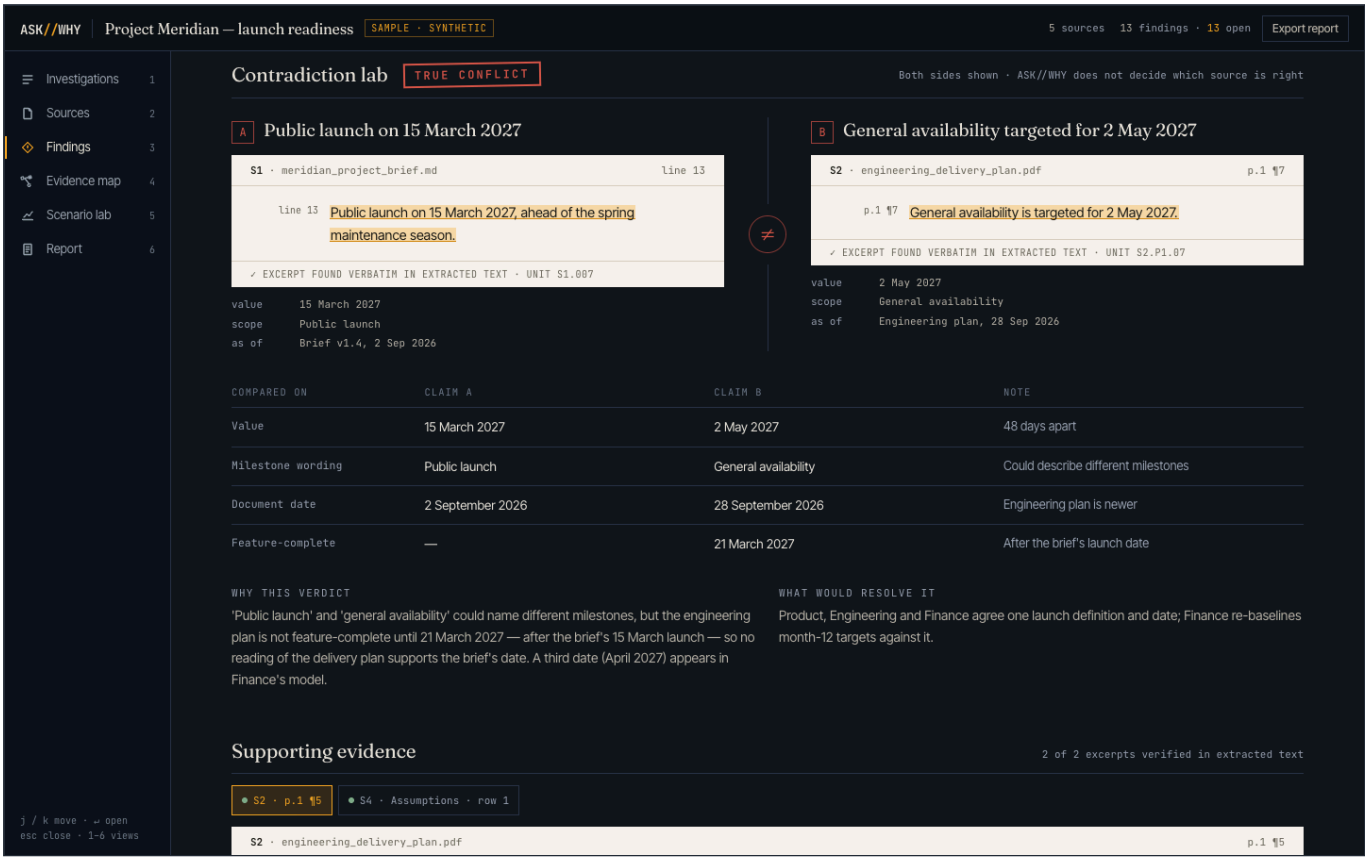
Landing page. The editorial headline, three starting points, and an intake form with live checks on each file.



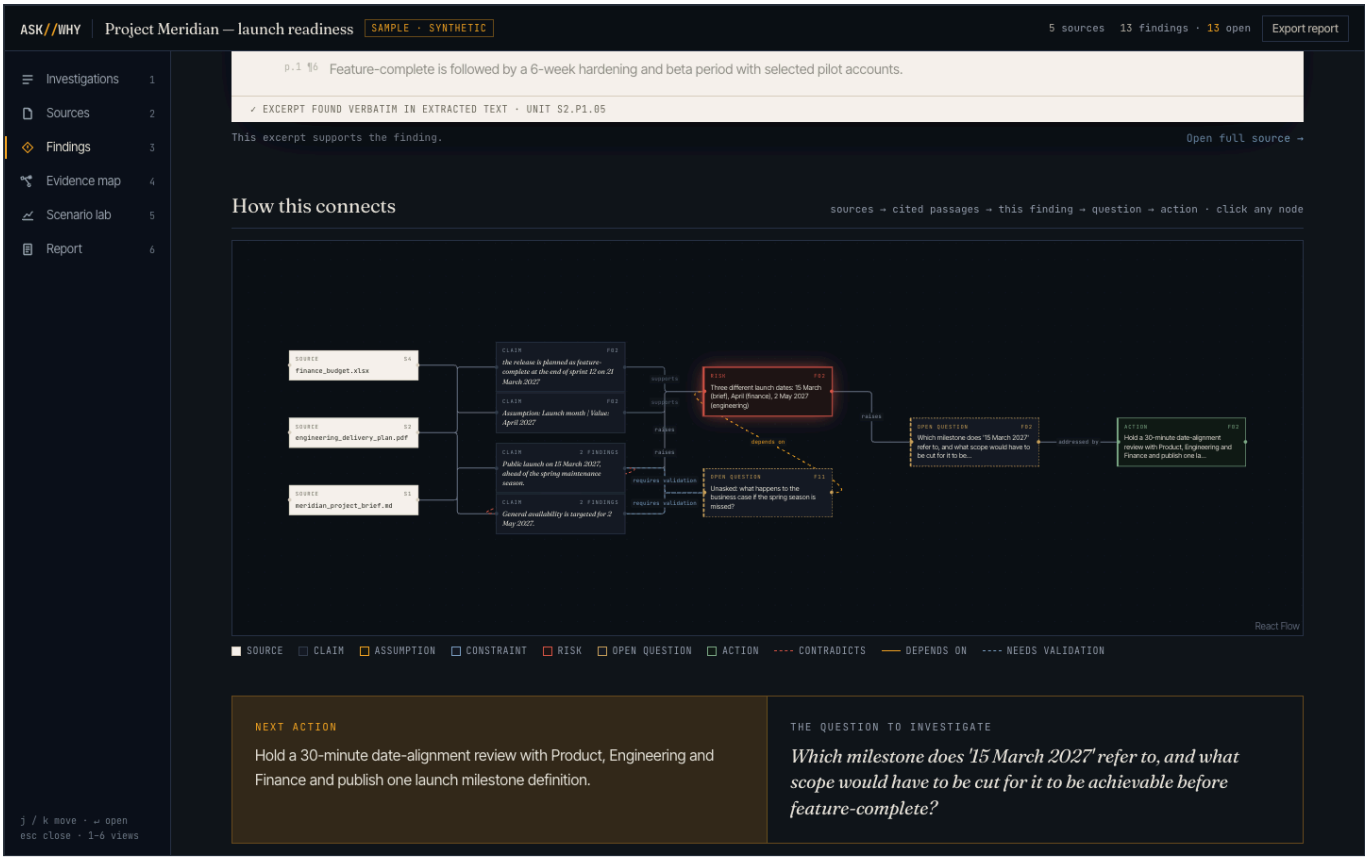
Findings overview. The framing question, verified-citation count, the top finding, an impact × confidence chart and the ranked list.



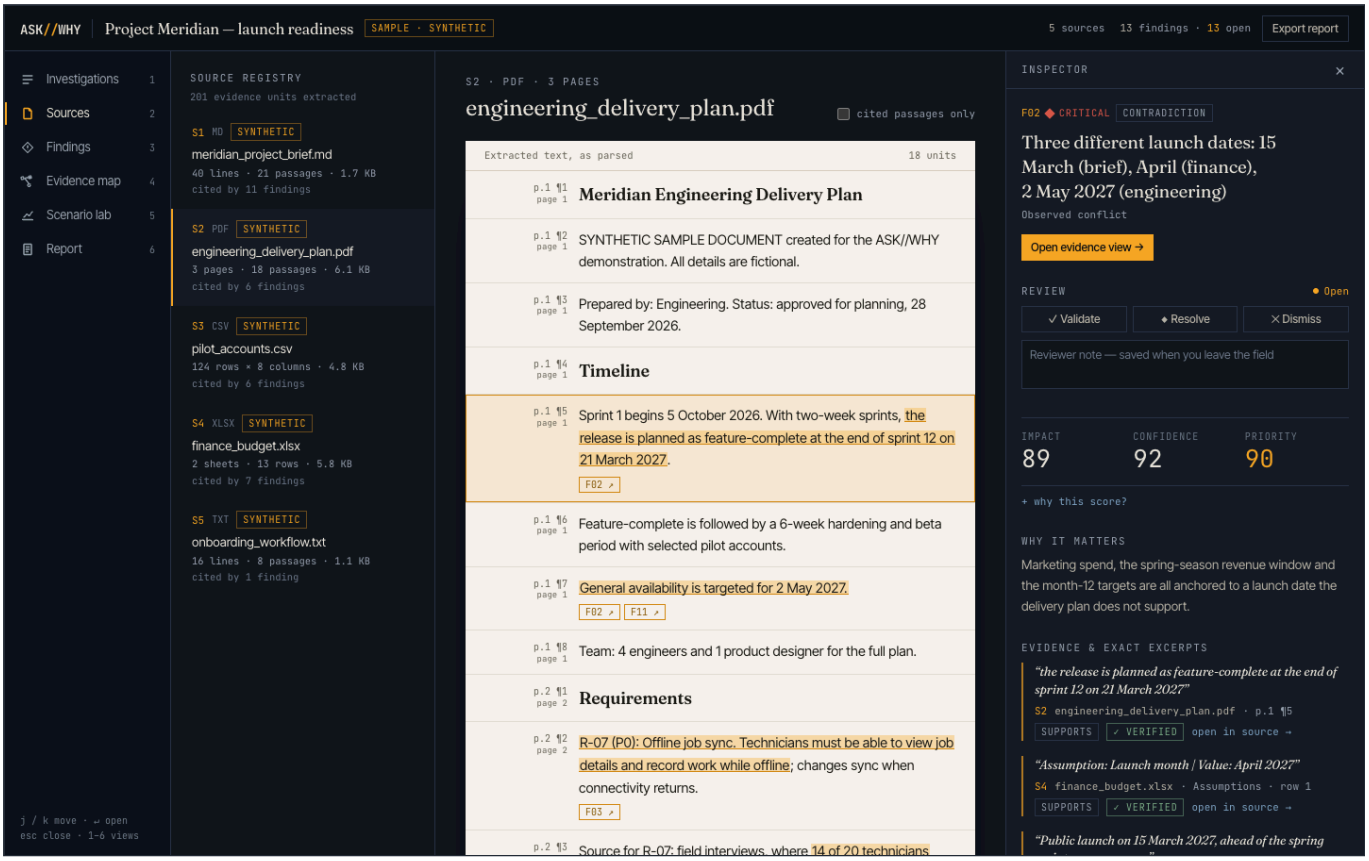
Evidence view (the signature interaction). A full-width dossier: the claim and why it matters on the left; review controls, scores and linked findings on the right.



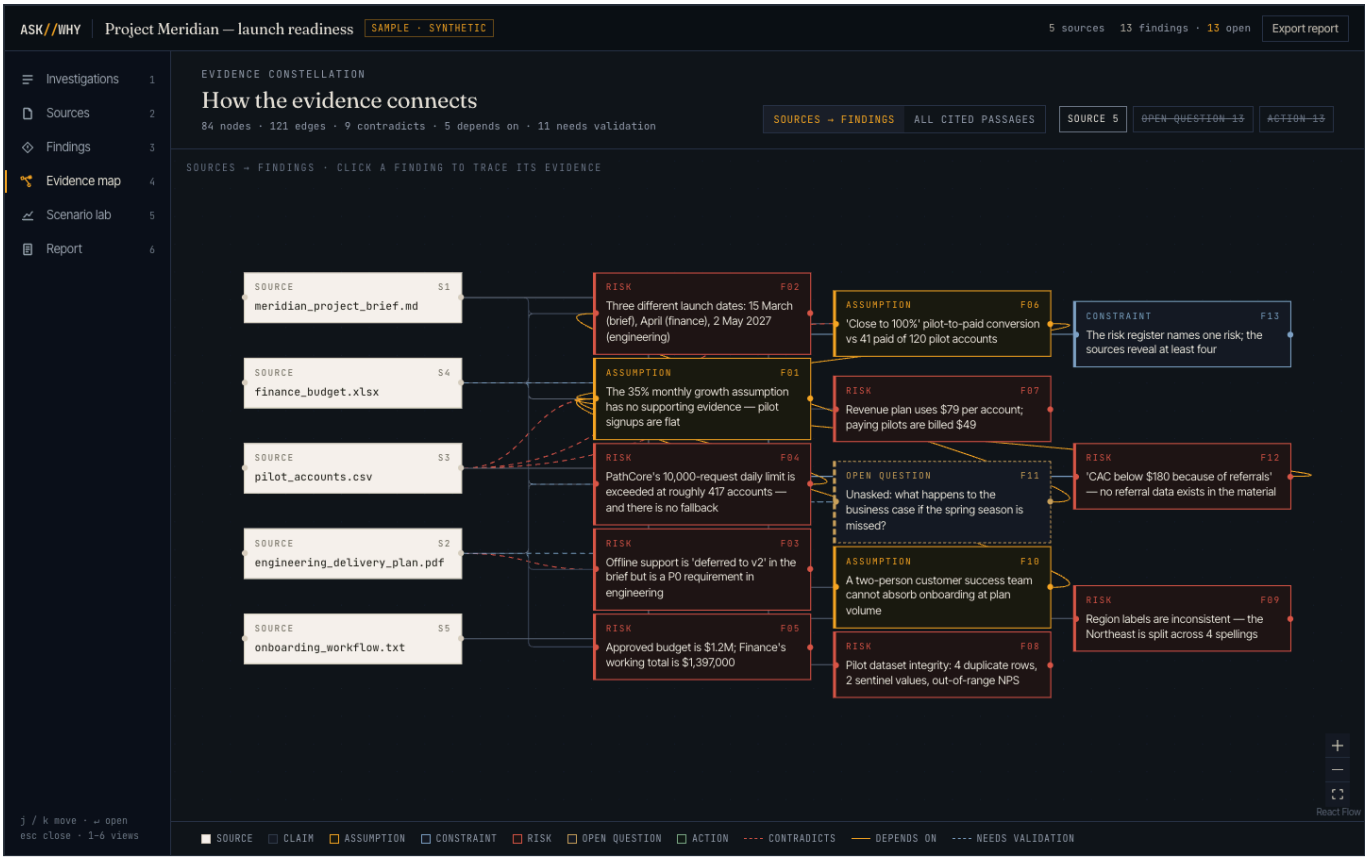
Contradiction Lab. Both claims on document sheets with the exact excerpt highlighted, a comparison table, the verdict and what would resolve it.



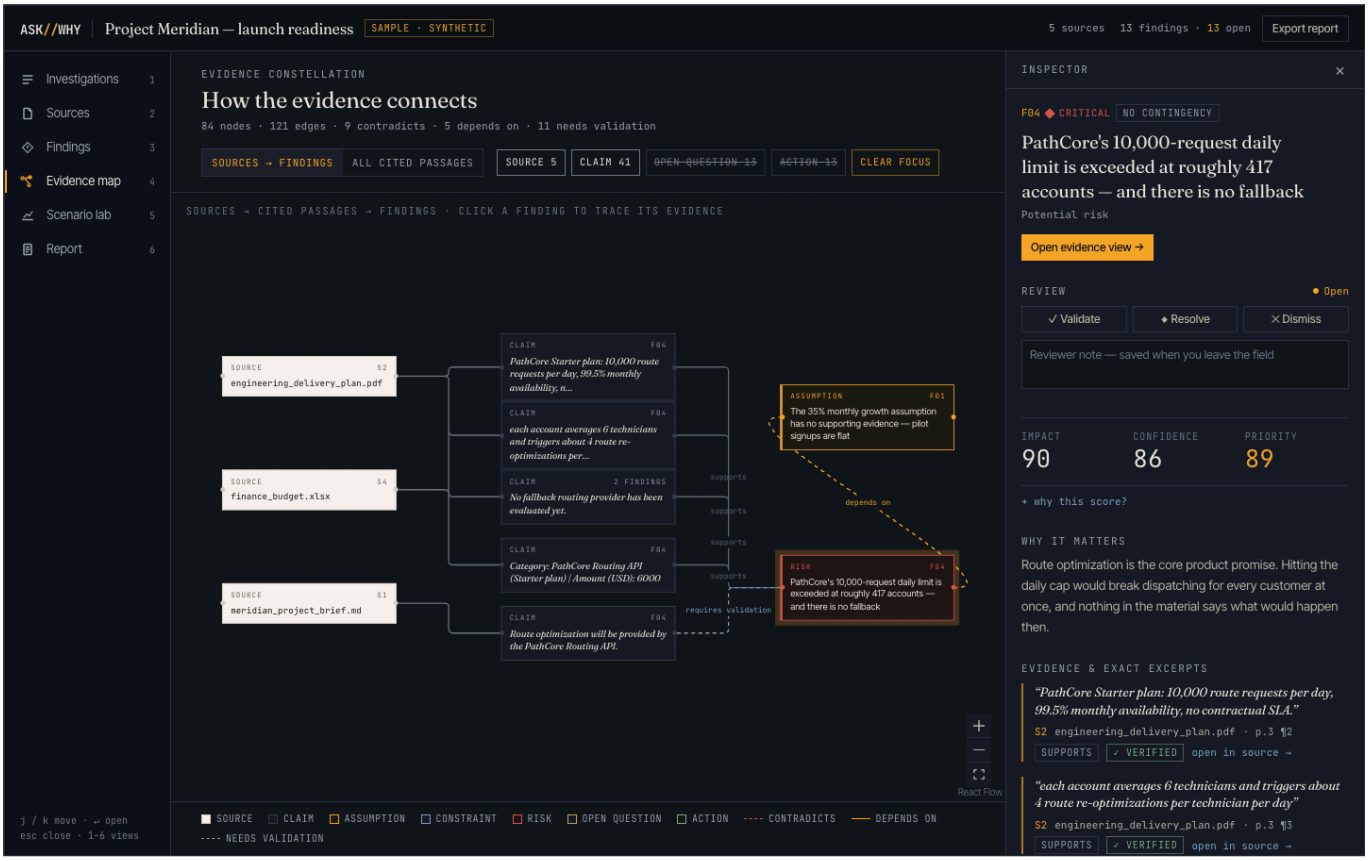
How this connects. The finding's own subgraph, from sources to the action, including the open question that depends on it.



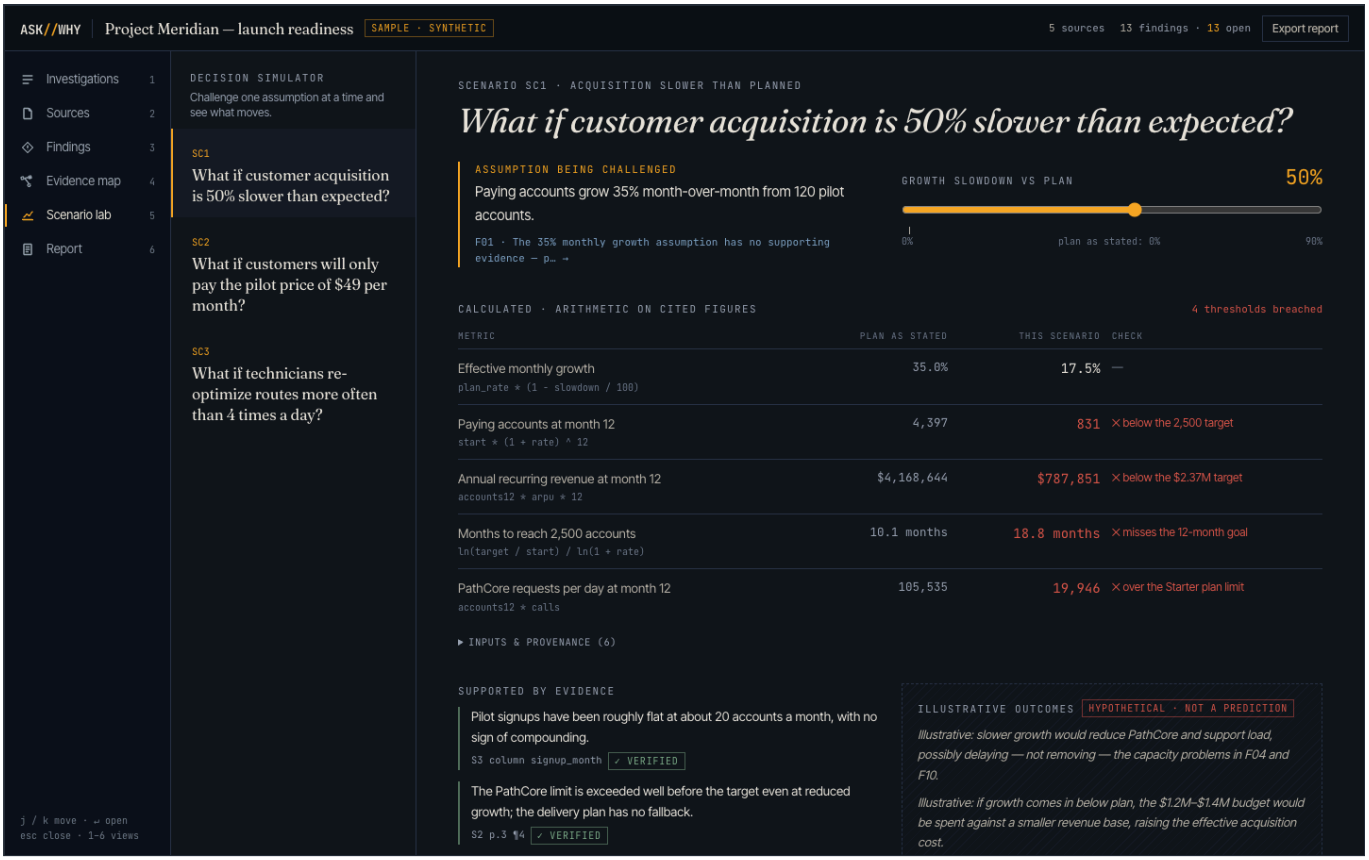
Sources reader. "Open full source" jumps to the exact PDF paragraph; cited passages are highlighted and link back to their findings.



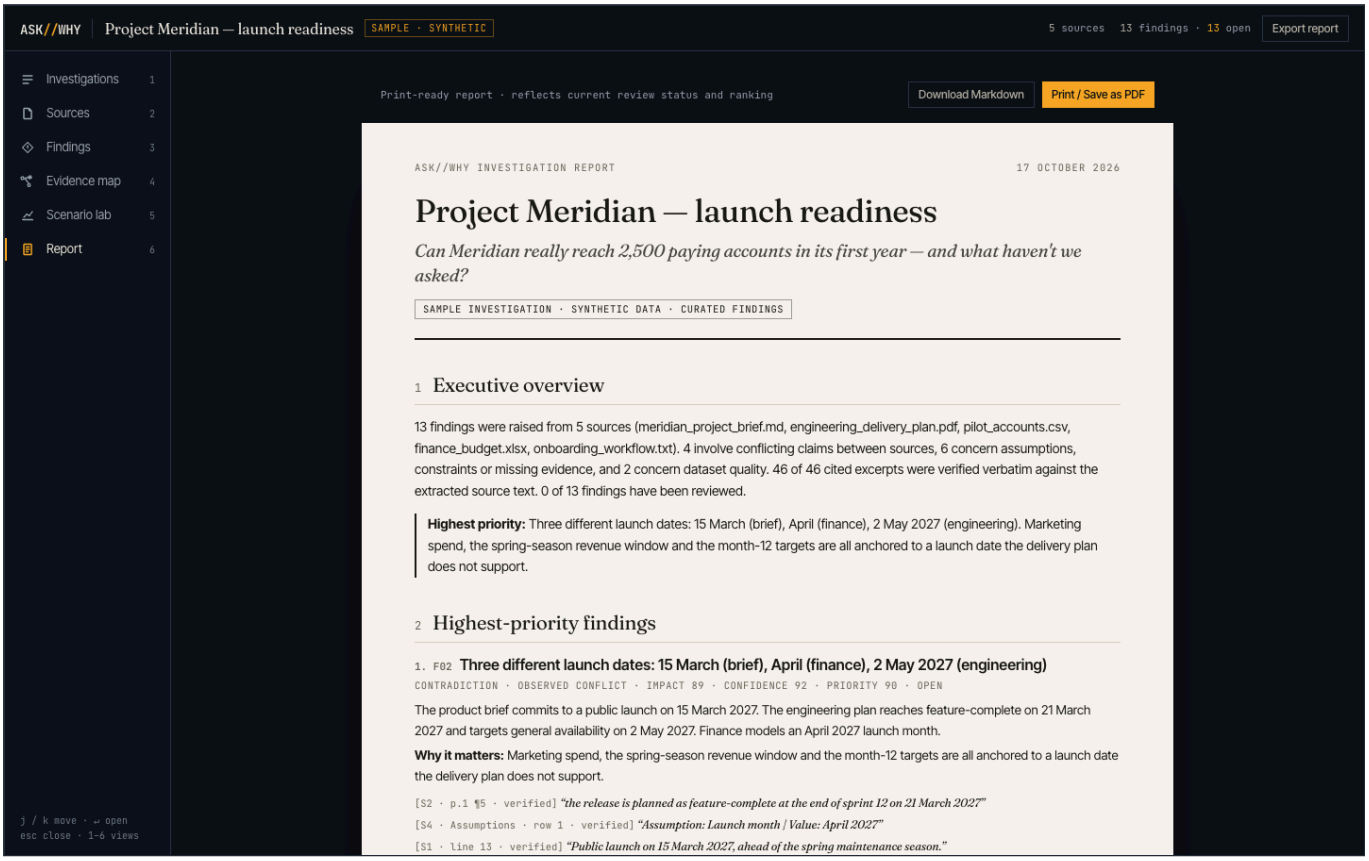
Evidence map, summary level. Sources → findings, with depends-on links. "All cited passages" shows the full graph.



Focused finding. Clicking F04 isolates its evidence chain; the amber *depends on* edge leads to the 35% growth assumption.



Scenario lab. Live arithmetic on cited figures with threshold checks; inputs list their sources; hypotheticals are clearly marked.



Investigation report. Executive overview, top findings with evidence, contradictions, open questions, actions and limitations.

05 Sample investigation: "Project Meridian"

Five synthetic files, all labelled as synthetic and read by the same parsers as real uploads: a product brief (MD), an engineering delivery plan (PDF), pilot account data (CSV), a finance workbook (XLSX) and an onboarding workflow (TXT). The curated findings cite sources by exact excerpt, and dataset figures are recomputed from the CSV every time the sample loads.

13	46/46	3	84
FINDINGS	CITATIONS VERIFIED	SCENARIOS	GRAPH NODES

REQUIRED CASE	EXAMPLE IN THE SAMPLE
Unsupported growth assumption	35% monthly growth assumed, but pilot signups are flat at about 20 a month
Conflicting requirement	Offline support "deferred to v2" in the brief, but a P0 requirement in engineering
Missing contingency plan	The PathCore API limit is exceeded at about 417 accounts, with no fallback provider
Dataset quality issue	4 duplicate rows, 9999 placeholder values, NPS of 11, inconsistent region labels
High-impact unanswered question	What happens to the business case if the spring launch window is missed?
Further findings	Three conflicting launch dates; $1.2M approved budget vs a $1,397,000 finance total; $79 planned price vs $49 actually billed; customer-success capacity gap

06 Testing & verification

CHECK	RESULT
Backend tests (pytest)	**42 passed**: parsing, profiling, rule-based findings, citation checking, scoring, API schemas, AI response conversion (no network), sample integrity
Frontend tests (vitest)	**7 passed**: formula evaluator, scenario arithmetic, excerpt locator, ranking
Production build	**Compiles** (TypeScript strict mode)
End-to-end demo	**Passed** in headless Chrome at 1440, 1280 and 390 px widths, with no console errors
Live upload of 5 files	**24 findings, 56/56 citations verified**, with live pipeline progress
Unsupported file (.exe)	**Rejected** with a clear message
Embedded-instruction test	**Ignored.** A DOCX containing "IGNORE ALL PREVIOUS INSTRUCTIONS" was treated as plain content

Bugs found and fixed during testing

- The highlighted excerpt was cut off after its first line.
- The graph was too small to read, and its zoom-to-finding never ran.
- The sample failed to open when the backend returned an error (it now falls back to the bundled copy).
- Shared links lost their place on page reload.
- A scenario showed "3500%" instead of 35%.

- The stored sample went stale after code changes; it now rebuilds on restart and keeps reviewer status.

Interface overhaul (second review round)

- **Evidence view** is now a full-width dossier. The duplicate inspector panel is gone, review and scoring sit beside the claim, and source passages get proper width.
- **Evidence map** has a readable "sources → findings" summary by default. Focusing a finding isolates its evidence chain instead of dimming 80 nodes.
- **Findings view:** tidier filter bar, ranking control moved into the list header, and chart labels no longer overlap.
- **Landing page:** the headline no longer wraps awkwardly, the format labels are cleaned up, and there is a clear "AI off" notice.
- **Readability:** higher contrast for small text and slightly larger labels.
- "Open evidence view" now works from the map and Sources screens, not only from Findings.

07 Limitations & things to know

> **AI not tested live.** No API key was available, so the AI layer was tested only with fake responses (errors, refusals, merged findings). To enable it, set `ANTHROPIC_API_KEY` in `backend/.env`. If an AI call fails, the analysis still completes with the rule-based engine.

- **Ports:** 5173 and 8000 were already used by another project on this machine, so ASK//WHY runs on **5174** (UI) and **8011** (API).
- **Rule-based engine:** it can miss conflicts that use different wording (e.g. "finish" vs "go-live"), and it reports the three launch dates as overlapping pairs.
- **PDFs:** only the text layer is read; scanned documents are not OCR'd.
- **Scores:** confidence is an evidence-strength rating, not a statistical probability.
- **Report export:** the in-app report exports through the browser's Print / Save as PDF.
- Findings support human judgement; important decisions still need human review.

08 How to run

STEP	COMMAND / ACTION
Start	`./start.sh` from the project root (installs dependencies on first run)
Open	`http://localhost:5174`, then click **Try sample investigation**
Enable AI	Copy `backend/.env.example` to `backend/.env` and set `ANTHROPIC_API_KEY`
Run tests	`cd backend && .venv/bin/python -m pytest -q` · `cd frontend && npx vitest run`
Documentation	`README.md`: setup, 3-minute demo script, scoring formulas, API and limitations

ASK//WHY: your biggest risk might be the question you never asked. · Build report generated 17 October 2026.